On the energy and momentum evolution in collisionless dark matter flow

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ABSTRACT

How do the energy and the momentum of collisionless dark matter grow in an expanding universe? We write the exact evolution equations of an N-body system in physical time for a pair potential $\propto r^n$: the cosmic energy equation $\mathrm{d}E/\mathrm{d}t + H(2K - nW) = 0$, the virial equation $\mathrm{d}G_y/\mathrm{d}t = 2K - nW$ for the radial momentum, and conservation of the physical angular momentum. If the kinetic energy grows as $K \propto a^\gamma$, the energy equation fixes the ratio of potential to kinetic energy and an effective exponent $n_e = 2K/W = -2(\gamma+1)/(\gamma+2)$ for $n = -1$. We test these relations with an Einstein–de Sitter simulation of 256^3 particles. (i) The cosmic energy equation holds to within 4.0 per cent at all redshifts $z \leqslant 5$. The kinetic energy grows as $a^{1.23}$ for $z \leqslant 1$, not as $a^{3/2}$ ($K \propto t$); with the local exponent 1.17 at $z = 0$ the energy equation predicts $n_e = -1.369$, and the simulation gives -1.365. (ii) For haloes of 100–300 particles the constants $\alpha_s^* = \sigma_v^2 r_g/(Gm_h) \approx 0.240$ and $\beta_s^* = Hr_g/\sigma_v \approx 0.106$ are nearly independent of mass; α_s^* is also independent of time, while $\beta_s^* \propto a^{-0.28}$. The spin parameter falls from 0.271 for halo pairs to 0.037 for haloes of more than 1000 particles. (iii) The radial momentum of spheres follows linear perturbation theory, $G_V = -(aH/3)\delta_V\langle|x|^2\rangle$ (correlation 0.83–0.97), and their angular momentum grows at second order, $\propto a^{3/2}$. The whole-box radial momentum of a periodic simulation depends on where the box is cut, so its value carries no physical information, and the integrals I_2 and I_4 of the velocity correlation do not converge for this spectrum.

Key words: dark matter – large-scale structure of Universe – methods: numerical – methods: analytical

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1 INTRODUCTION

The question. Dark matter in an expanding universe keeps gaining kinetic energy: haloes form, merge and become virialized, and the peculiar velocities grow. How fast does the energy grow, how is it shared between kinetic and potential energy, and what happens to the radial and angular momentum of the system and of its haloes? These are the questions of this paper.

What is known. The energy of a self-gravitating system in an expanding background obeys the cosmic energy equation of Irvine [1] and Layzer [2] [see also 3, 4]. It is one equation for two unknowns, the kinetic energy K and the potential energy W, much like the energy equation of freely decaying turbulence, which needs an extra relation (Loitsyansky's integral) to give the decay laws [5, 6]. In the cascade picture of dark matter flow, haloes pass mass to larger haloes and kinetic energy is produced on small scales at a rate ε_u [7, 8]. One way to close the energy equation combines the two-body collapse model [9] with a constant rate of energy production; it gives $K \propto t$ and an effective potential exponent $n_e = -10/7$ (Section 3.5). By analogy with turbulence one may also look for dimensionless constants of the energy and momentum of haloes and for integral constants I_m of the velocity correlation.

What this paper does. We (1) write the exact equations for energy, radial momentum and angular momentum in physical time and for any power-law potential (Section 3); (2) test them, and the closure, against the simulation (Section 4); (3) measure the halo constants at eight redshifts (Sections 5 and 6); and (4) test the large-scale radial and angular momentum against perturbation theory and examine

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Table 1. Parameters of the simulation.

Run	Ω_0	Λ	h	Γ	σ_8	L (Mpc/h)	N	m_p ($M_\odot/h$)	$l_{\rm soft}$ (kpc/h)
SCDM1	1.0	0.0	0.5	0.5	0.51	239.5	256^3	2.27×10^{11}	36

Table 2. Notation (specific quantities, per unit mass). $\langle\cdot\rangle$ is a particle average.

K	peculiar kinetic energy $\langle\lvert u\rvert^2\rangle/2$ [km^2 s^{-2}]
W	potential energy of the density fluctuations [km^2 s^{-2}]
E	$K+W$
γ	growth exponent of K: $\mathrm{d}\ln K/\mathrm{d}\ln a$
n_e	effective potential exponent $2K/W$
G_y, H_y	radial momentum $\langle r\cdot u\rangle$ and angular momentum $\langle r\times u\rangle$ (physical) [km s^{-1} Mpc/h]
G, H	the same with comoving x ($G_y=aG$, $H_y=aH$)
r_g, σ_v^2	physical rms radius of a halo about its centre of mass; one-dimensional velocity dispersion in the halo
Φ_h, E_h	specific potential energy of a halo; $E_h=\frac{3}{2}\sigma_v^2+\Phi_h$
G_h, $\lvert H_h\rvert$	radial and angular momentum of the particles of a halo about its centre of mass (physical)
α_s^*, β_s^*	$\sigma_v^2 r_g/(Gm_h)$ and Hr_g/σ_v
n_s^*, z_s^*	$3\sigma_v^2/\Phi_h$ and E_h/σ_v^2
τ_s^*, η_s^*	$-G_h a/(\sigma_v r_g)$ and $\lvert H_h\rvert/(\sigma_v r_g a^{1/2})$
λ_p	spin parameter $\lvert H_h\rvert\lvert E_h\rvert^{1/2}/(Gm_h)$
α_T	G^2/T with $T=M:M$, $M_{ij}=\langle x_i u_j\rangle$

whether the integrals I_2 and I_4 exist (Section 7). Section 8 ranks the evidence and lists tests; Section 9 concludes.

The answers in brief. The equations hold in the simulation. The kinetic energy grows more slowly than t, and the measured effective exponent ($n_e\approx-1.37$) is exactly what the energy equation requires for the measured growth rate. Two halo constants are nearly universal. Sub-volumes follow linear theory for the radial momentum and second-order theory for the angular momentum, while whole-box values of a periodic simulation are not physical.

2 SIMULATION AND MEASURED QUANTITIES

We use the Virgo consortium SCDM run [10, 11]: an Einstein–de Sitter (EdS) universe, $\Omega_0=1$, with the parameters of Table 1. Its particle data and halo catalogues (friends-of-friends with a comoving linking length of 0.2 Mpc/h, i.e. $b=0.214$ of the mean interparticle separation) were used in the earlier papers of this series; the derived statistics are public [12, 13, 14]. In EdS, $a=(t/t_0)^{2/3}$, $H=H_0a^{-3/2}$ and $t_0=2/(3H_0)=13.04$ Gyr for $h=0.5$. Comoving positions x are in Mpc/h; physical positions are $r=ax$. The peculiar velocity is $u=a\,\mathrm{d}x/\mathrm{d}t$ (km s^{-1}), and $u(a)$ is its one-dimensional rms value ($u_0=354.61$ km s^{-1} at $z=0$).

All energies and momenta are *specific* (per unit mass) and averaged over particles. The quantities used throughout are collected in Table 2. The kinetic energy is $K=\langle\lvert u\rvert^2\rangle/2=3u^2/2$. The potential energy W is that of the density fluctuations (the mean density drives the expansion and is removed). We measure it in two independent ways: from the potential of every particle (a sum over all particles minus its value for a uniform distribution) and from the two-point correlation function, $W=-(3H_0^2/4a)\int\xi(x)x\,\mathrm{d}x$ [3]. At $z\leqslant1$ the two agree to within 2 per cent (ratio 1.018 at $z=1$ and 1.006 at $z=0$); at $z=5$, where $\lvert W\rvert$ is small and the correlation is noisy, the ratio is 1.31.

3 EXACT EVOLUTION EQUATIONS

3.1 Equations of motion

Consider N particles of mass m_p that interact through a pair potential $-G_n m_p r^n$ ($n=-1$ is Newtonian gravity, $G_{-1}=G$). The physical peculiar acceleration of particle i is $g_i=nG_nm_p\sum_{j\neq i}(r_i-r_j)\lvert r_i-r_j\rvert^{n-2}$. In comoving coordinates the equation of motion is

$$\frac{\mathrm{d}^2x_i}{\mathrm{d}t^2}+2H\frac{\mathrm{d}x_i}{\mathrm{d}t}=\frac{g_i}{a}=\frac{nG_nm_p}{a^{2-n}}\sum_{j\neq i}\frac{x_i-x_j}{\lvert x_i-x_j\rvert^{2-n}}. \tag{1}$$

The power of a follows from $r=ax$: the force falls as a^{n-1} and one more factor a^{-1} converts a physical into a comoving acceleration. For gravity the factor is the familiar a^{-3} [3]. In terms of the peculiar velocity, equation (1) reads

$$\frac{\mathrm{d}u_i}{\mathrm{d}t}=-Hu_i+g_i,\qquad\frac{\mathrm{d}r_i}{\mathrm{d}t}=Hr_i+u_i. \tag{2}$$

The first term on the right is the Hubble drag on peculiar motion.

3.2 Energy: the cosmic energy equation

The potential energy is $W=\frac{1}{N}\sum_i\phi_i$ with $\phi_i=-\frac{1}{2}G_nm_p\sum_{j\neq i}\lvert r_i-r_j\rvert^n$. Its time derivative has two parts: at fixed comoving positions $W\propto a^n$, which gives nHW, and the motion relative to the background gives $-\langle u\cdot g\rangle$. With equation (2), $\mathrm{d}K/\mathrm{d}t=-2HK+\langle u\cdot g\rangle$. Adding the two,

$$\frac{\mathrm{d}E}{\mathrm{d}t}+H\,(2K-nW)=0,\qquad E=K+W. \tag{3}$$

For gravity this is the Layzer–Irvine equation, $\mathrm{d}E/\mathrm{d}t+H(2K+W)=0$. It says that the expansion removes energy at the rate $H(2K+W)$: the total energy is not conserved, and it is constant only for a system in virial equilibrium ($2K+W=0$). We checked equation (3) by integrating a small N-body system in an EdS background for $n=-1$ and $n=-\frac{1}{2}$; it holds to the accuracy of the integrator (10^{-11}).

3.3 Power-law growth fixes the energy ratio

The energy equation has one relation for two unknowns. Suppose, as the simulation suggests (Section 4), that K and W grow as the same power of a, $K\propto W\propto a^\gamma$. Then $\mathrm{d}E/\mathrm{d}t=\gamma HE$ and equation (3) gives $\gamma(K+W)+2K-nW=0$, i.e.

$$\frac{W}{K}=-\frac{\gamma+2}{\gamma-n},\qquad n_e\equiv\frac{2K}{W}=-\frac{2(\gamma-n)}{\gamma+2},\qquad\frac{E}{K}=-\frac{n+2}{\gamma-n}. \tag{4}$$

The effective exponent n_e is the exponent for which the system would satisfy the static virial theorem $2K=n_eW$. For gravity,

$$n_e=-\frac{2(\gamma+1)}{\gamma+2},\qquad\gamma=-\frac{2(1+n_e)}{2+n_e},\qquad\frac{2K+W}{K}=\frac{\gamma}{\gamma+1}. \tag{5}$$

A static system ($\gamma=0$) has $n_e=-1$ and obeys the usual virial theorem. A growing system ($\gamma>0$) has $-2<n_e<-1$: it holds less potential energy per unit kinetic energy than a virialized one, because the kinetic energy keeps being produced. Table 3 lists examples. Linear theory has $\gamma=1$ ($u^2\propto a$) and $n_e=-4/3$. A system whose kinetic energy grows as $K\propto t$ has $\gamma=3/2$ and $n_e=-10/7$. Equation (4) was checked numerically: integrating equation (3) with $K\propto a^\gamma$ from $W=0$ converges to the ratio W/K of equation (4) for any γ and n.

Table 3. Steady (power-law) states of the cosmic energy equation for gravity, equation (5). The values $\gamma = 1.17$ and 1.23 are the measured local exponent at $z = 0$ and the fit for $z \leqslant 1$ (Section 4).

γ	0	1	1.17	1.23	3/2	3	∞
n_e	-1	$-4/3$	-1.369	-1.381	$-10/7$	$-8/5$	-2
W/K	-2	$-3/2$	-1.461	-1.448	$-7/5$	$-5/4$	-1
E/K	-1	$-1/2$	-0.461	-0.448	$-2/5$	$-1/4$	0

3.4 Radial and angular momentum

Define the radial momentum (the virial quantity) $G_y = \langle r \cdot u \rangle$ and the angular momentum $H_y = \langle r \times u \rangle$, both in physical coordinates. With equation (2), the Hubble terms cancel, and for an isolated system of particles with pair forces (Euler's theorem for a potential of degree n, $\langle r \cdot g \rangle = nW$, and $\langle r \times g \rangle = 0$)

$$\frac{\mathrm{d}G_y}{\mathrm{d}t} = 2K - nW, \qquad \frac{\mathrm{d}H_y}{\mathrm{d}t} = 0. \tag{6}$$

The first is the virial theorem in an expanding background: G_y grows when $2K > nW$. The second says that the physical angular momentum is conserved; the comoving $H = H_y/a$ therefore decays as a^{-1}. For gravity, combining equations (3) and (6) gives the compact form

$$\frac{\mathrm{d}E}{\mathrm{d}t} = -H\frac{\mathrm{d}G_y}{\mathrm{d}t}, \tag{7}$$

so the energy lost to the expansion equals H times the growth of the radial momentum (verified to 2×10^{-11} with the N-body test). Equations (6) and (7) apply to an isolated system. In a periodic simulation the sums $\langle r \cdot g \rangle$ and $\langle r \times g \rangle$ depend on where the box is cut, and Section 7 shows that whole-box values of G and H do too.

3.5 A transformed time and the two-body closure

For gravity it is convenient to use a transformed time s with $\mathrm{d}s = a^{-3/2}\mathrm{d}t$. In comoving coordinates and time s, equation (1) becomes $\mathrm{d}^2x_i/\mathrm{d}s^2 + \frac{1}{2}H_0\,\mathrm{d}x_i/\mathrm{d}s = Gm_p\sum_j(x_j - x_i)/|x_j - x_i|^3$: a system with a fixed damping rate $H_0/2$ in a static background. (For $n \neq -1$ a factor a^{1+n} remains on the right, so the static form is special to gravity.) With $K_s = aK$ and $W_s = aW$ the energy equation takes the form of a damped oscillator, $\mathrm{d}(K_s + W_s)/\mathrm{d}s + H_0K_s = 0$, equivalent to equation (3). The two-body collapse model [9] suggests exponential solutions in s, $K_s = \alpha e^{-ks}$ and $W_s = \beta e^{-ks}$, with $k = H_0/(1 + \beta/\alpha)$; in physical time these are the power laws of Section 3.3 with $n_e = 2\alpha/\beta$ and $\gamma = -(2 + \beta/\alpha)/(1 + \beta/\alpha)$. The closure therefore fixes the *form* of the solution but not the value of γ. The value $\gamma = 3/2$ ($K \propto t$, $n_e = -10/7$) follows only if one also assumes that the kinetic energy is produced at a constant rate.

4 SYSTEM ENERGY IN THE SIMULATION

Figure 1 shows K, $-W$ (from the potentials and from the correlation function), $|E|$ and $2K + W$ for the ten snapshots. For $z \leqslant 1$ the energies grow as $K \propto a^{1.23}$ and $-W \propto a^{1.26}$. The lower panel tests the cosmic energy equation directly: we evaluate $a\,\mathrm{d}E/\mathrm{d}a + 2K + W$ with finite differences in $\ln a$ and divide by $2K + |W|$. The residual is at most 4.0 per cent ($z \leqslant 5$), at most 1.5 per cent for $z \leqslant 0.5$, and 0.4 per cent at $z = 0$; the residuals are consistent with the coarse spacing of the snapshots. The simulation obeys equation (3).

Figure 2 (upper panel) shows the local growth exponent $\gamma = \mathrm{d}\ln K/\mathrm{d}\ln a$. It starts at the linear-theory value (1.04 at $z = 10$), peaks at 1.32 at $a = 0.40$, and declines to 1.17 at $z = 0$. It never reaches

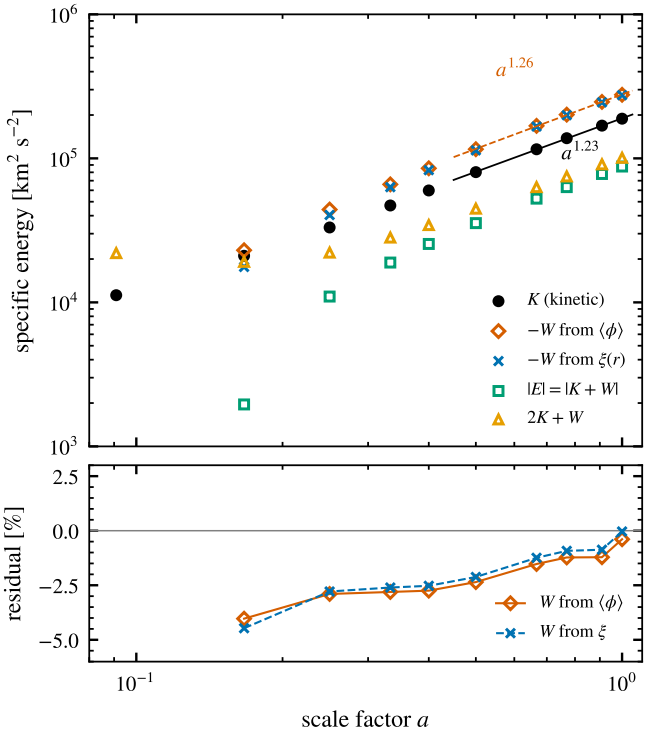


Figure 1. The energies grow as power laws and obey the cosmic energy equation. Upper: kinetic energy K, potential energy $-W$ from the particle potentials (diamonds) and from the correlation function (crosses), $|E|$ and $2K + W$; lines are fits for $z \leqslant 1$. Lower: relative residual of the cosmic energy equation, $(a\,\mathrm{d}E/\mathrm{d}a + 2K + W)/(2K + |W|)$.

3/2, so K does not grow as t: with the fit for $z \leqslant 1$, $K \propto t^{0.82}$, and locally at $z = 0$, $K \propto t^{0.78}$.

The lower panel tests equation (5). The measured $n_e = 2K/W$ approaches -1.365 at $z = 0$ (-1.374 with W from the correlation function). The steady-state prediction with the measured local γ, $-2(\gamma + 1)/(\gamma + 2)$, is -1.369 at $z = 0$ and -1.393 at $z = 1$ (measured -1.386); for $z \leqslant 2$ the two differ by at most 0.04. At $z \geqslant 3$ the system has not reached a steady state (measured $n_e = -1.50$ at $z = 3$ and -1.83 at $z = 5$, where $W/K = -1.09$ instead of $-3/2$). So the effective exponent near -1.37 is not evidence for $K \propto t$; it is what the energy equation requires for $\gamma \approx 1.2$. Conversely, $n_e = -1.38$ inserted in equation (5) gives $\gamma = 1.23$, not 3/2.

The energy cascade rate. The series uses the rate $\varepsilon_u = -\frac{3}{2}u_0^2/t_0 = -\frac{9}{4}u_0^2H_0 \approx -4.6 \times 10^{-7}\ \mathrm{m^2\,s^{-3}}$ as the characteristic rate at which kinetic energy is produced on small scales [8]. This value equals $-\mathrm{d}K/\mathrm{d}t$ only if $K \propto t$. With the measured local exponent, $\mathrm{d}K/\mathrm{d}t = \gamma HK = 3.6 \times 10^{-7}\ \mathrm{m^2\,s^{-3}}$ at $z = 0$, i.e. 0.78 of $|\varepsilon_u|$. The scale of ε_u is therefore right to within about 20 per cent, but the present simulation does not show a constant rate.

5 HALO ENERGY: TWO NEARLY UNIVERSAL CONSTANTS

5.1 Definitions and a simple model

For every halo we measure its mass $m_h = n_pm_p$ (n_p particles), the physical rms radius r_g about its centre of mass, the one-dimensional velocity dispersion σ_v^2 about the halo velocity, and the specific

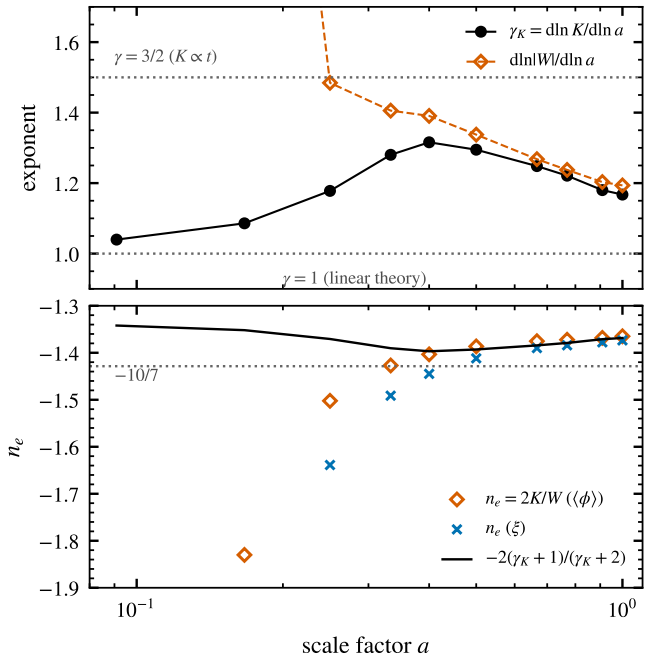


Figure 2. The effective exponent follows from the measured growth rate. Upper: local exponents $\mathrm{d}\ln K/\mathrm{d}\ln a$ and $\mathrm{d}\ln|W|/\mathrm{d}\ln a$. Lower: $n_e = 2K/W$ from the potentials (diamonds) and from the correlation function (crosses), and the steady-state relation $n_e = -2(\gamma+1)/(\gamma+2)$ evaluated with the measured γ (line). The dotted line is $-10/7$, the value for $K \propto t$.

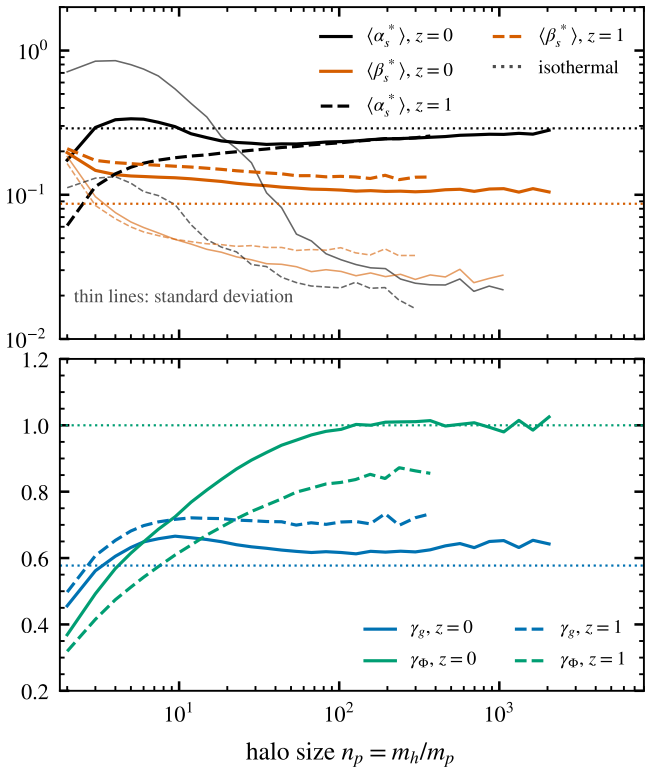


Figure 3. Two nearly mass-independent halo constants. Upper: mean (thick) and standard deviation (thin) of α_s^* and β_s^* against halo size at $z = 0$ (solid) and $z = 1$ (dashed); dotted lines are the isothermal values. Lower: $\gamma_g = r_g/r_\Delta$ and $\gamma_\Phi = -\Phi_h r_\Delta/(Gm_h)$. Halo-weighted means in logarithmic bins with at least 20 haloes.

potential energy Φ_h of its particles. At $z = 0$ the catalogue has 1,674,583 haloes with 10,290,483 particles (the largest has 6326). Two dimensionless numbers combine these quantities:

$$\alpha_s^* = \frac{\sigma_v^2 r_g}{Gm_h}, \qquad \beta_s^* = \frac{Hr_g}{\sigma_v}. \tag{8}$$

α_s^* is a virial ratio (kinetic over gravitational energy scale), and β_s^* compares the expansion rate with the internal crossing rate σ_v/r_g.

A simple model shows what to expect. Let a halo have the radius r_Δ inside which its mean density is $\Delta_c = 18\pi^2$ times the background density, $m_h = \Delta_c\bar\rho\,\frac{4\pi}{3}r_\Delta^3$ with $\bar\rho = 3H^2/(8\pi G)$, so $r_\Delta = (2Gm_h/\Delta_c H^2)^{1/3}$. Write $r_g = \gamma_g r_\Delta$, $\Phi_h = -\gamma_\Phi Gm_h/r_\Delta$ and $3\sigma_v^2 = -\gamma_v\Phi_h$ ($\gamma_v = 1$ for virial equilibrium). Then

$$\beta_s^* = \left(\frac{6\gamma_g^2}{\gamma_\Phi\gamma_v\Delta_c}\right)^{1/2}, \qquad \alpha_s^* = \frac{\gamma_\Phi\gamma_v\gamma_g}{3}. \tag{9}$$

For a truncated power-law profile $\rho \propto r^{-m}$, $\gamma_g = [(3-m)/(5-m)]^{1/2}$ and $\gamma_\Phi = (3-m)/(5-2m)$; for an NFW profile [15] with concentration $c = 4$–10, $\gamma_g = 0.56$–0.49 and $\gamma_\Phi = 0.97$–1.25 (by quadrature over the profile). The singular isothermal sphere ($m = 2$: $\gamma_g = 1/\sqrt{3}$, $\gamma_\Phi = 1$) with $\gamma_v = 1.5$ gives $\alpha_s^* = 0.289$ and $\beta_s^* = 0.087$; we call these the isothermal estimates.

5.2 Measurements

Figure 3 shows the halo-weighted means and the standard deviations of α_s^* and β_s^* against halo size at $z = 0$ and $z = 1$, together with $\gamma_g = r_g/r_\Delta$ and γ_Φ measured halo by halo. Both constants scatter widely for small haloes and settle to nearly constant values above $n_p \approx 100$. For $100 \leqslant n_p \leqslant 300$ at $z = 0$ we find $\alpha_s^* = 0.240$ and $\beta_s^* = 0.106$, i.e. about 20 per cent below and above the isothermal values. The reason is visible in the lower panel: the measured $\gamma_g \approx 0.617$ is larger than $1/\sqrt{3}$, and the measured virial ratio $\gamma_v = -n_s^* \approx 1.19$ (Section 5.3) is smaller than 1.5. Inserting the measured γ_g, γ_Φ and γ_v in equation (9) reproduces the measured β_s^* to within 3.8 per cent and α_s^* to within 4.2 per cent at all redshifts ($\beta_s^* = 0.104$ predicted at $z = 0$). The small difference comes from averaging products versus products of averages.

Figure 4 follows the constants of the haloes with $100 \leqslant n_p \leqslant 300$ through time. α_s^* stays between 0.236 and 0.240 from $z = 1.5$ to 0 ($\propto a^{0.01}$): it is a universal constant. β_s^* is not: it falls from 0.137 at $z = 1.5$ to 0.106 at $z = 0$, $\beta_s^* \propto a^{-0.28}$. Equation (9) explains the trend: at higher redshift the haloes are less concentrated relative to r_Δ (larger γ_g, smaller γ_Φ; Fig. 3), while the product $\gamma_\Phi\gamma_g$ that sets α_s^* hardly changes.

5.3 Virial ratio and energy ratio of haloes

Two further ratios describe the energy balance of a halo: the halo exponent $n_s^* = 2K_h/\Phi_h = 3\sigma_v^2/\Phi_h$ (the analogue of n_e; $n_s^* = -1$ in virial equilibrium) and $z_s^* = E_h/\sigma_v^2 = \frac{3}{2} + 3/n_s^*$. Figure 5 shows both against halo size. Haloes with more than about 30 particles have $n_s^* \approx -1.2$ (-1.192 for $100 \leqslant n_p \leqslant 300$ and -1.270 above 1000 particles) and $z_s^* \approx -0.9$ to -1.0 (-1.044 and -0.871). Like the whole system, haloes hold less potential energy per unit kinetic energy than a static virialized object, as expected for objects that keep accreting mass [16]. Halo pairs ($n_p = 2$) are far from equilibrium at $z = 0$

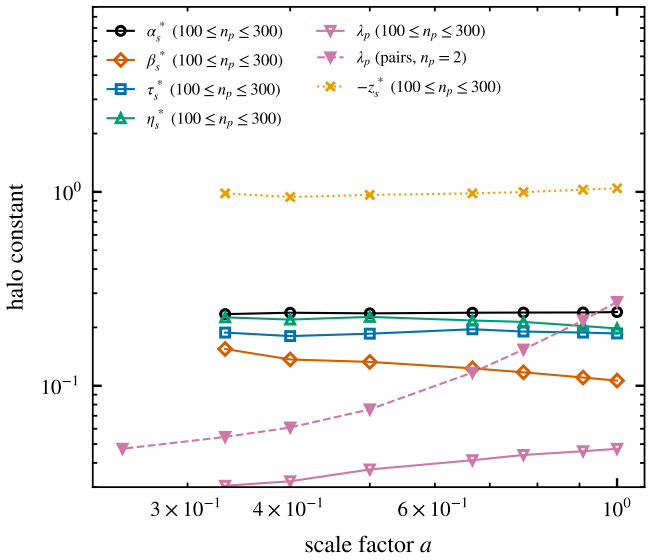


Figure 4. Which halo constants change with time. Means over haloes with $100 \leqslant n_p \leqslant 300$ against scale factor: α_s^*, β_s^*, τ_s^*, η_s^*, $-z_s^*$ and the spin parameter λ_p; dashed: λ_p of halo pairs ($n_p = 2$).

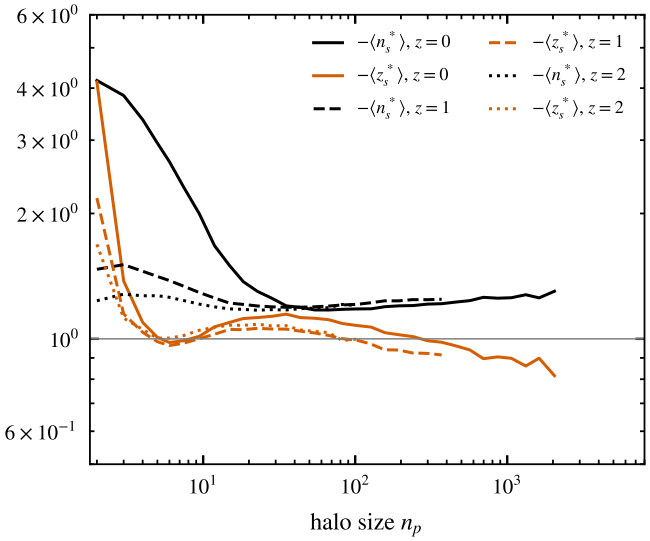


Figure 5. Haloes are not in static virial equilibrium. Halo exponent $-n_s^* = -3\sigma_v^2/\Phi_h$ and energy ratio $-z_s^* = -E_h/\sigma_v^2$ against halo size at $z = 0$, 1 and 2. The grey line marks 1.

($n_s^* = -4.173$): most of them are tightly bound pairs selected by the linking length.

6 HALO MOMENTUM AND SPIN

6.1 Radial and angular momentum of haloes

For each halo we compute the radial momentum $G_h = \langle r' \cdot u' \rangle$ and the angular momentum $|H_h| = |\langle r' \times u' \rangle|$ of its particles about the centre of mass (primes: relative to the halo). Figure 6 shows both against halo size. $G_h < 0$ for all sizes: on average particles fall inwards.

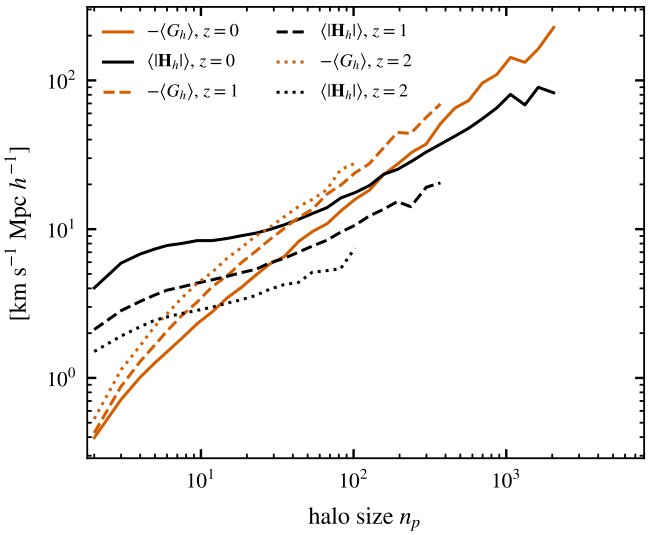


Figure 6. Rotation dominates small haloes, infall dominates the largest. Mean radial momentum $-G_h$ and angular momentum $|H_h|$ of halo particles about the centre of mass (physical units) against halo size at $z = 0$, 1 and 2.

Rotation dominates small haloes, and the radial infall is larger than the rotation in haloes with more than about 300 particles.

Natural units for both are $\sigma_v r_g$. We define

$$\tau_s^* = -\frac{G_h\, a}{\sigma_v r_g}, \qquad \eta_s^* = \frac{|H_h|}{\sigma_v r_g\, a^{1/2}}. \tag{10}$$

The factors of a follow from a simple model in which $G_h \propto a^{-1}$ and $|H_h| \propto a^{1/2}$ at fixed $\sigma_v r_g$. The data support them: for $100 \leqslant n_p \leqslant 300$, $\tau_s^* \propto a^{0.04}$ and $\eta_s^* \propto a^{-0.13}$, while without the factors the ratios would change as $a^{-0.96}$ and $a^{0.37}$ (Fig. 4). At $z = 0$, $\tau_s^* = 0.186$ and $\eta_s^* = 0.197$ for these haloes.

For two-body haloes the two-body collapse model gives exact values: at the critical ratio $\beta_{s2} = 1/(3\pi)$ of radial to circular velocity, $\eta_s^* = \sqrt{3}$, $\tau_s^* = \sqrt{3}/(3\pi)$, $\beta_s^* = \sqrt{3}/(3\pi)$ and $\alpha_s^* = 1/24$ [9]. Halo pairs approach these values at early times: at $z = 3$ they have $\alpha_s^* = 0.049$, $\beta_s^* = 0.167$ and $\eta_s^* = 1.954$ (model: 0.042, 0.184 and 1.73), but at $z = 0$ the population is dominated by old, tightly bound pairs ($\alpha_s^* = 0.174$, $\eta_s^* = 1.296$).

6.2 Spin parameter

The spin parameter [17, 18] in specific quantities is $\lambda_p = |H_h||E_h|^{1/2}/(Gm_h)$. With the definitions above it factorizes exactly, halo by halo,

$$\lambda_p = a^{1/2}\, \alpha_s^*\, \eta_s^*\, |z_s^*|^{1/2}, \tag{11}$$

(checked on all 1,674,582 haloes at $z = 0$ to rounding error). Because α_s^* and z_s^* change little with mass for large haloes, the mass dependence of the spin comes from η_s^*. Figure 7 shows τ_s^*, η_s^* and λ_p against halo size. The mean spin falls from 0.271 for pairs to 0.047 for $100 \leqslant n_p \leqslant 300$ and 0.037 above 1000 particles. The distribution of λ_p for haloes with more than 100 particles peaks at 0.042 (median 0.040), close to the values of about 0.035 found for simulated haloes [19]. For pairs the distribution peaks at 0.075 (median 0.062); the mean is pulled up by a tail of nearly unbound pairs. The two-body collapse value for equal masses, $\lambda_p = \sqrt{2}/16 = 0.088$, lies near the peak.

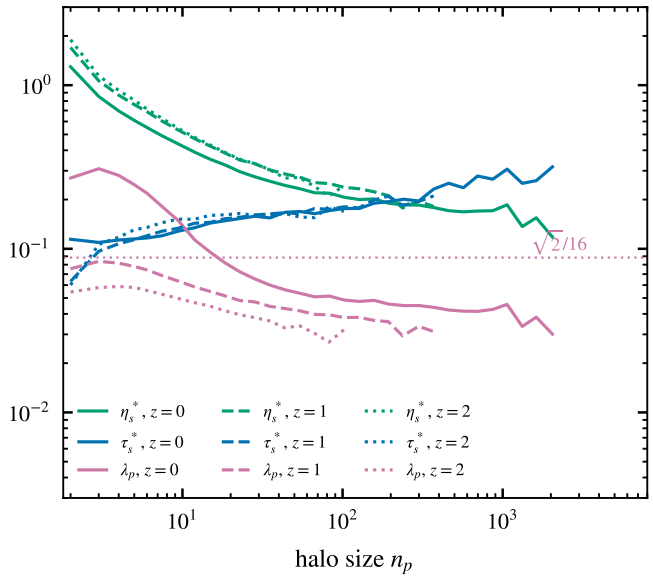


Figure 7. Spin falls with halo mass. Momentum coefficients η_s^* and τ_s^* (equation 10) and spin parameter λ_p against halo size at $z = 0$, 1 and 2. The dotted line is the two-body collapse value $\sqrt{2}/16$.

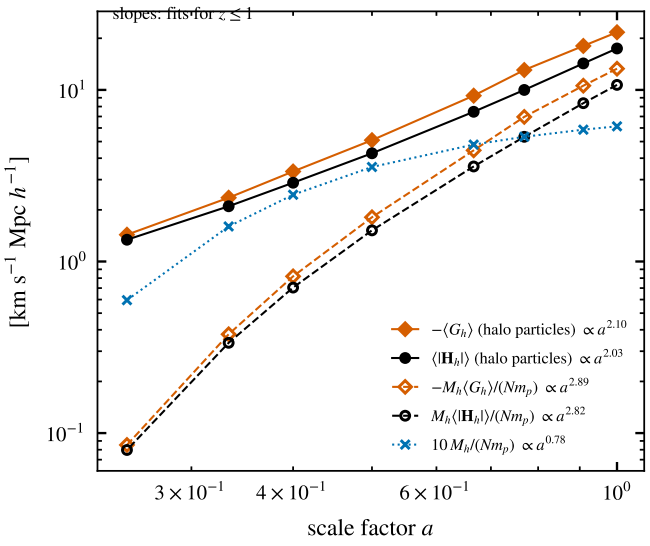


Figure 8. Momentum of the halo subsystem grows as about a^2 per halo particle. Mean radial and angular momentum per halo particle (solid), totals divided by the number of particles in the box (dashed), and ten times the halo mass fraction, against scale factor; exponents are fits for $z \leqslant 1$.

The spin grows with time at all sizes (Fig. 4): for $z \leqslant 1$ the mean spin of pairs grows as $a^{1.84}$ and that of haloes with $10 \leqslant n_p \leqslant 30$ as $a^{0.83}$; for $100 \leqslant n_p \leqslant 300$, $\lambda_p \propto a^{0.41}$ since $z = 1.5$. Equation (11) with constant α_s^*, η_s^* and z_s^* would give $a^{1/2}$; the faster growth of small haloes comes from their α_s^* and z_s^*, which change with time.

6.3 The halo subsystem

Figure 8 follows the momentum of all halo particles together. The mean radial and angular momentum per halo particle grow as $|\langle G_h \rangle| \propto a^{2.10}$ and $\langle |H_h| \rangle \propto a^{2.03}$ (fits for $z \leqslant 1$; at $z = 0$, $\langle G_h \rangle = -21.7$ and $\langle |H_h| \rangle = 17.4\,\mathrm{km\,s^{-1}\,Mpc}/h$). The totals per particle of the box grow faster, $\propto a^{2.89}$ and $a^{2.82}$, because the halo mass fraction itself grows ($\propto a^{0.78}$; 0.61 at $z = 0$). Near $z = 0$ the local exponents are 2.40, 2.58 and 0.47.

7 LARGE-SCALE RADIAL AND ANGULAR MOMENTUM

7.1 Whole-box values depend on where the box is cut

The simplest large-scale measures are the radial momentum $G = \langle x \cdot u \rangle$, the angular momentum $H = \langle x \times u \rangle$ and $T = M : M$ ($M_{ij} = \langle x_i u_j \rangle$) of the whole simulation box, with x measured from the centre of mass. For the box as stored (its original origin) they grow as $G \propto a^{0.49}$ and $|H| \propto a^{1.57}$, and $\alpha_T = G^2/T \approx 2$, a value that might be read as collapse into filaments.

The box is periodic, so the position of its origin is arbitrary. Figure 9 repeats the measurement for 16 origins. At $z = 0$, G ranges from -6958 to $4505\,\mathrm{km\,s^{-1}\,Mpc}/h$ (the original origin gives -4580); the mean is 363 ± 890, consistent with zero; and α_T ranges from 0.01 to 2.18 (the original origin gives 2.18, nearly the largest). The time dependence, in contrast, is the same for every origin: for the 11 origins with $|G| > 2000$ at $z = 0$, $G(z = 1)/G(z = 0) = 0.700 \pm 0.038$ and $G(z = 10)/G(z = 0) = 0.298 \pm 0.026$, against $a^{1/2} = 0.707$ and 0.302.

Linear theory explains both facts. The peculiar velocity of the growing mode is the gradient of a velocity potential, $u = -\nabla\psi$, with $\psi \propto aHD \propto a^{1/2}$ for $D \propto a$. For the cube V cut out of the periodic box, the divergence theorem gives $G = -V^{-1}\int x \cdot \nabla\psi\,\mathrm{d}^3x = -(L/2V)\oint \psi\,\mathrm{d}S$, because $\int \psi\,\mathrm{d}^3x = 0$ and $x \cdot n = L/2$ on every face. The radial momentum is thus set by the potential on the six faces where the box is cut: it grows as $a^{1/2}$ but its value and sign depend on the cut. For the angular momentum, $x \times n$ takes opposite values on opposite faces while ψ is equal there (periodicity), so the first-order term vanishes; what remains is of second order, $\propto \delta u \propto a^{3/2}$, which is the growth found (median exponent 1.53 over the origins). The value $\alpha_T \approx 2$ is therefore a property of one cut, not of the dynamics, and it says nothing about filamentary collapse.

7.2 Spheres follow perturbation theory

Spheres inside the box have no such ambiguity. We measured G_V, H_V and T_V in 120 spheres with random centres and radii $R = 5$, 10, 20, 40 and 60 Mpc$/h$ at all snapshots, with positions measured from the centre of mass of each sphere. If the mean divergence inside a sphere is $\bar\theta = \overline{\nabla \cdot u}$ and the flow is linear, $u = Ax$ with $\mathrm{tr}A = \bar\theta$, then $G_V = \langle |x|^2 \rangle \bar\theta/3$ exactly (the inertia tensor of a sphere is isotropic). With linear theory $\bar\theta = -aHf\delta_V$ ($f = 1$ in EdS), where δ_V is the overdensity of the sphere,

$$G_V = -\frac{aHf}{3}\,\delta_V\,\langle |x|^2 \rangle. \tag{12}$$

Figure 10 compares the measured G_V with equation (12) at $z = 5$ and $z = 0$. The correlation is 0.83–0.97 for all radii and redshifts, and the least-squares slope is 1.03–1.16 for $R \geqslant 10\,\mathrm{Mpc}/h$. Only the smallest spheres at $z = 0$ (rms $\delta_V = 0.92$) depart from linear theory (slope 0.86).

For spheres $\alpha_T = \langle G_V^2 \rangle / \langle T_V \rangle = 0.80$–$1.04$ (mean 0.94). This is the value for an isotropic Gaussian velocity field: in linear theory M_{ij} is proportional to the Hessian $\partial_i\partial_j\psi$ averaged over the sphere, whose isotropic Gaussian statistics give $\langle (\mathrm{tr}M)^2 \rangle = \langle M : M \rangle$, i.e. $\alpha_T = 1$ (Appendix A; a Monte Carlo test on a Gaussian field gives 1.03).

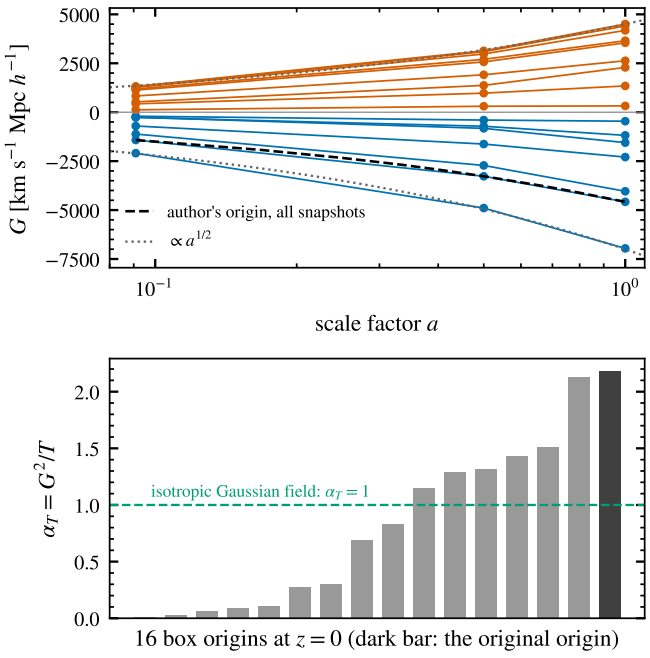


Figure 9. The whole-box radial momentum is not physical. Upper: $G = \langle x \cdot u \rangle$ of the periodic box against scale factor for 16 positions of the box origin (red: $G > 0$ at $z = 0$, blue: $G < 0$); dashed: the original origin at all snapshots; dotted: $\propto a^{1/2}$. Lower: $\alpha_T = G^2/T$ at $z = 0$ for the same origins.

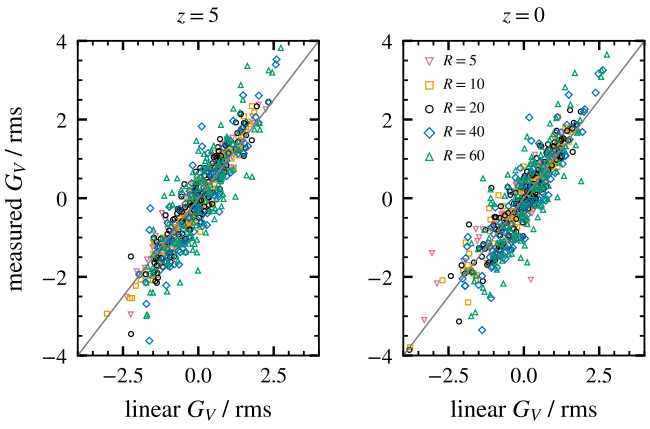


Figure 10. Sub-volumes obey linear theory. Radial momentum G_V of 120 random spheres against the linear prediction, equation (12), at $z = 5$ and $z = 0$, both divided by the rms of the prediction for each radius; the line is equality.

The values 3, 2 and 1 for collapse to a point, a filament and a plane hold for a single coherent flow; a random field mixes them.

Figure 11 shows the growth of the rms radial and angular momentum of the spheres. In the linear regime ($z \geqslant 2$) the rms G_V grows as $a^{0.46}$–$a^{0.49}$, as equation (12) requires ($aH\delta \propto a^{1/2}$). The rms $|H_V|$ grows as $a^{1.47}$–$a^{1.58}$. For an irrotational flow the volume-weighted angular momentum of a sphere vanishes ($x \times n = 0$ on its surface), so the particle-weighted $H_V = \langle (1+\delta)\, x \times u \rangle$ is of second order, $\propto \delta u \propto a^{3/2}$; on a Gaussian field the density-weighted H_V grows with the square of the amplitude (exponent 2.06; volume-weighted residue 0.001 of G_V). The ratio $|H_V|/G_V$ therefore grows as a: from

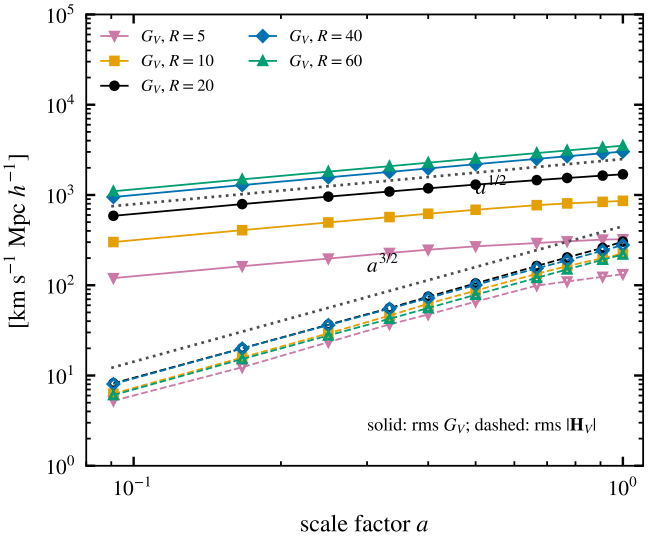


Figure 11. Radial momentum is first order, angular momentum second order. Rms comoving radial momentum G_V (solid) and angular momentum $|H_V|$ (dashed) of spheres of radius R (Mpc/h) against scale factor; dotted lines $\propto a^{1/2}$ and $a^{3/2}$.

0.006 at $z = 10$ to 0.062 at $z = 0$ for $R = 60\,\mathrm{Mpc}/h$, and to 0.41 for $R = 5\,\mathrm{Mpc}/h$. This is why angular momentum is small on large scales. (The angular momentum of a fixed set of particles grows by tidal torques, $\propto t$ in physical units, Peebles 17, White 20; our spheres are fixed comoving volumes, not fixed sets of particles.)

7.3 Integral constants

In turbulence the decay laws follow from the integrals of the velocity correlation $R_2(r) = \langle u \cdot u' \rangle$,

$$I_m = \int R_2(r)\, r^{m-2}\, \mathrm{d}^3 r = \int_0^\infty 4\pi R_2 r^m \,\mathrm{d}r, \tag{13}$$

Saffman's I_2 [21] and Loitsyansky's I_4 [6, 22]. With the velocity spectrum $E_u(k) = \pi^{-1} \int R_2 kr \sin(kr)\,\mathrm{d}r$ [23], $I_m = 4\pi^2 (-1)^{1+m/2} m^{-1}\, \partial^m E_u / \partial k^m |_{k=0}$, so the integrals describe the largest scales. For a closed volume with zero net momentum, $I_4 = -2 \lim_{V\to\infty} V \langle T \rangle$, and the identity

$$|H|^2 = 2T - M : \overline{(x_i x_j u_k)_{,k}} + (M : I)\, \bar{\theta} \tag{14}$$

(I the inertia tensor, overbars volume averages) reduces to the Landau–Loitsyansky relation $\langle |H|^2 \rangle = 2\langle T \rangle$ for an incompressible flow without surface flux. In dark matter flow the divergence term does not vanish and, as shown above, $|H| \ll G$ on large scales, so T is set by the radial momentum.

Do the integrals exist? $I_2 = \lim_{V\to\infty} V \langle \bar{u}_V^2 \rangle$ requires the variance of the mean velocity of a volume to fall faster than $1/V$. In linear theory the velocity spectrum is $P(k)/k^2$; for $P \propto k^{n_s}$ at small k, $V\langle \bar{u}_V^2 \rangle \propto R^{2-n_s}$, which grows without bound for the CDM value $n_s = 1$, and I_4 diverges faster still. Figure 12 shows the partial integrals $I_2(r_{\max})$ and $I_4(r_{\max})$ of the measured $R_2(r)$ [24]. Both level off between 60 and 90 Mpc/h (changes of 1.7 and 2.4 per cent) but change again near half the box size (I_2 by 11.6 per cent from 90 to 119.75 Mpc/h; I_4 by a factor 2.17 from 60 to 119.75 Mpc/h); over the whole periodic box $\int R_2\, \mathrm{d}^3 r = 0$ by construction. The partial integrals therefore measure the largest modes of this box rather than constants of the flow. What the data do support is their growth with

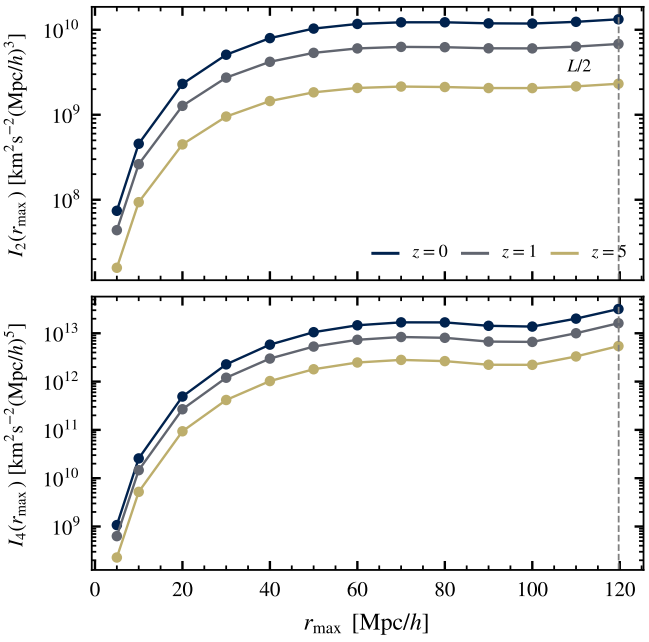


Figure 12. The integrals of the velocity correlation do not converge. Partial integrals $I_2(r_{\rm max})$ and $I_4(r_{\rm max})$ of $R_2(r)$ at $z = 0$, 1 and 5; the dashed line marks half the box size.

time: $I_2(r \leqslant 90) \propto a^{0.98}$ and $I_4(r \leqslant 90) \propto a^{1.02}$, as expected from $u^2 \propto a$ on these scales.

8 DISCUSSION

Evidence, ranked. (1) The equations of Section 3 are exact for an isolated system; they were verified to 10^{-11} by direct integration and need no assumption about haloes. (2) The simulation obeys the cosmic energy equation to within 4.0 per cent, and its effective exponent follows the steady-state relation with the measured growth rate to within 0.04 for $z \leqslant 2$. (3) α_s^* is the same for all large haloes at all redshifts (≈ 0.240); β_s^* and the spin depend on time. (4) Subvolumes follow linear theory for the radial momentum and second-order theory for the angular momentum.

What the data rule out. The simulation does not show $K \propto t$: the kinetic energy grows as $a^{1.2}$, and the effective exponent $n_e \approx -1.37$ is what the energy equation predicts for that rate, not the value $-10/7$ of $K \propto t$. The halo constants are about 20 per cent from the isothermal estimates, and β_s^* changes with time. The whole-box $\alpha_T \approx 2$ is set by where the box is cut, and the integrals I_2 and I_4 do not exist for a CDM spectrum, so they cannot serve as constants of motion.

Tests. The relations are falsifiable with any simulation. (a) In any run, n_e must track $-2(\gamma+1)/(\gamma+2)$ with the local growth exponent of K; a run that reaches $K \propto t$ must show $n_e \to -10/7$. (b) Spheres of any radius must show $\alpha_T \approx 1$ and $|H_V|/G_V \propto a$ while linear theory holds. (c) α_s^* should be universal in other cosmologies, while β_s^* should follow the concentration of haloes relative to r_Δ through equation (9).

Open problems. Why does the growth exponent peak at about 1.3 near $z \approx 1.5$ and then decline? Longer or ΛCDM runs are needed to see whether it approaches 3/2. The halo constants were measured with friends-of-friends haloes; spherical-overdensity haloes may shift γ_g and hence β_s^*. The angular momentum of fixed sets of particles (tidal torques) was not measured here. The constant energy-cascade rate used in the series is a characteristic value: in this run the measured $\mathrm{d}K/\mathrm{d}t$ at $z = 0$ is 0.78 of it (Section 4), and it should be read as the scale of the energy production rather than as a measured constant.

9 CONCLUSIONS

We derived, in physical time and for a pair potential $\propto r^n$, the exact equations for the energy ($\mathrm{d}E/\mathrm{d}t + H(2K - nW) = 0$), the radial momentum ($\mathrm{d}G_y/\mathrm{d}t = 2K - nW$) and the angular momentum (conserved) of an N-body system in an expanding universe, and their power-law steady states. Tested on an EdS simulation:

(i) The cosmic energy equation holds to within 4.0 per cent. The kinetic energy grows as $a^{1.23}$ for $z \leqslant 1$ (local exponent 1.17 at $z = 0$), not as t, and the effective exponent $n_e = -1.365$ is what the energy equation requires for that rate.

(ii) Large haloes have $\alpha_s^* \approx 0.240$ at all redshifts and $\beta_s^* \approx 0.106$ at $z = 0$, rising as $a^{-0.28}$ to higher redshift; a three-parameter profile model reproduces both from the measured γ_g, γ_Φ and γ_v. Haloes have $n_s^* \approx -1.2$.

(iii) The radial and angular momentum coefficients τ_s^* and η_s^* are nearly time independent with the scalings $G_h \propto a^{-1}$, $|H_h| \propto a^{1/2}$; the spin factorizes exactly as $\lambda_p = a^{1/2}\alpha_s^*\eta_s^*|z_s^*|^{1/2}$ and falls from 0.271 for pairs to 0.037 for the largest haloes.

(iv) On large scales the radial momentum follows linear theory and the angular momentum second-order theory; whole-box values of a periodic simulation depend on where the box is cut, and the velocity-correlation integrals I_2 and I_4 do not converge for a CDM spectrum.

DATA AVAILABILITY

The halo-based and correlation-based statistics of the simulation are on Zenodo [12, 13], with presentation slides [25]; all files are listed in Xu [14]. The reduced data, the code that produces every figure and number of this paper, and the checks are in the accompanying reproducible package (DOI assigned on deposit).

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APPENDIX A: WHY SPHERES HAVE $\alpha_T = 1$

In linear theory $u = -\nabla\psi$ with ψ a homogeneous, isotropic Gaussian field. Inside a sphere small compared with the coherence length, $u_j \approx u_j(0) - \psi_{,jk}(0)x_k$, so $M_{ij} = \langle x_i u_j\rangle = -\psi_{,jk}\langle x_i x_k\rangle = -\frac{1}{3}\langle|x|^2\rangle\psi_{,ij}$ (the term $u_j(0)\langle x_i\rangle$ vanishes about the centre of mass). For an isotropic Gaussian field the second derivatives have the covariance $\langle\psi_{,ij}\psi_{,kl}\rangle = \frac{\sigma^2}{15}(\delta_{ij}\delta_{kl}+\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk})$ with $\sigma^2 = \langle(\nabla^2\psi)^2\rangle$. Hence $\langle(\psi_{,ii})^2\rangle = \frac{\sigma^2}{15}(9+3+3) = \sigma^2$ and $\langle\psi_{,ij}\psi_{,ij}\rangle = \frac{\sigma^2}{15}(3+9+3) = \sigma^2$, so $\langle G^2\rangle = \langle T\rangle$ and $\alpha_T = 1$. A uniform contraction ($\psi_{,ij} \propto \delta_{ij}$) would give 3; the Gaussian field has shear as well as compression, and on average the two contribute so that $\alpha_T = 1$.